

L1.7a Implicit Differentiation

Work in your group. Check your answers against the Quarto tonight.

Implicit curves and level sets

1. For each relation, say whether you **can solve it for y** — and if you can, whether you would *want* to.

(a) $y^3 - x = 4$

(b) $x^2 + y^2 = 25$

(c) $x^3 + y^3 = \frac{9}{2}xy$

(d) $\sin y + x^2y = 4$

2. Let $S(x, y) = x^2 + y^2$, so the circle $x^2 + y^2 = 25$ is the curve $S = 25$. Start at $(3, 4)$.

- (a) Take the step $dx = -0.1$, $dy = -0.1$. Find dS , and find the level you land on. **Are you still on the circle?**

- (b) Now take $dx = 0.1$ and find the dy that makes $dS = 0$. Then write down $\frac{dy}{dx}$. **What have you just found?**

Implicit differentiation

3. Differentiate each with respect to x . **These are terms, not equations — nothing to solve.**

$$\frac{d}{dx}(y^3)$$

$$\frac{d}{dx}(xy)$$

$$\frac{d}{dx}(\sin y)$$

$$\frac{d}{dx}(x^2y)$$

Why does a y pick up a y' and an x does not?

4. Take $\frac{d}{dx}$ of both sides. **Stop there — do not solve for y' .**

(a) $x^2 + y^2 = 25$

(b) $x^3 + y^3 = \frac{9}{2}xy$

(c) $y + x \cos y = x^2$

5. Now finish the job — find y' .

(a) $x^2 + y^2 = 25$

(b) $x^2 + xy + y^2 = 3$

(c) $y + x \cos y = x^2$

Stop. Before item 6: your answer to 5(a) has a y in it. Is that finished, or not?

Tangent lines to an implicit curve

6. Find the equation of the tangent line.

(a) $x^2 + y^2 = 25$ at $(3, 4)$ (compare with the entry ticket)

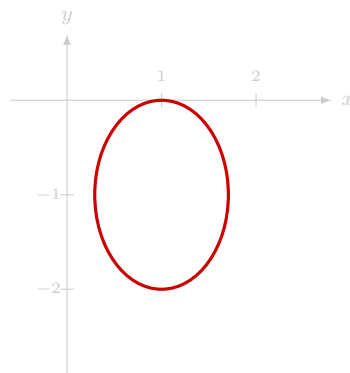
(b) $x^3 + y^3 = \frac{9}{2}xy$ at $(2, 1)$ (check the point is on the curve first)

Horizontal and vertical tangents

7. Use your y' from item 5(b). On $x^2 + xy + y^2 = 3$, find **every** point with a horizontal tangent and **every** point with a vertical tangent.

8. **Stretch.** Where on $x^2 + y^2 = 25$ is the tangent line **parallel to** $y = x$?

9. **BOARDS** The ellipse $2x^2 + y^2 - 4x + 2y + 2 = 0$ is graphed below. Find **every** horizontal tangent and **every** vertical tangent.



Two of the four need factoring that is not obvious. Use Loh.